ICT305 Topics in IT

Topic 3
Reinforcement Learning

CIHE Australia

Materials

- David Silver, “Lectures on Reinforcement Learning,”

Reinforcement Learning

- So far …

Supervised Learning

→ Cat

Classification

Unsupervised Learning

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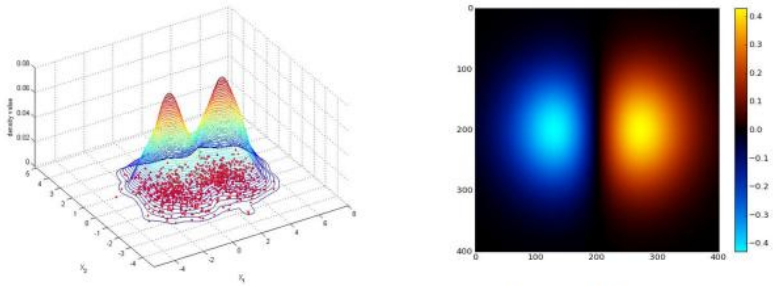
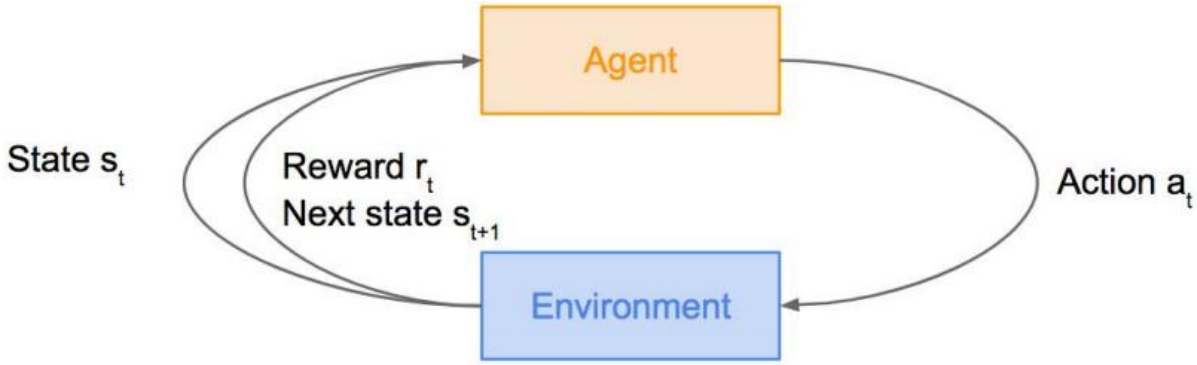
1-d density estimation



2-d density estimation

Reinforcement Learning

- **Problem**: an agent interacting with an environment, which provides numeric *rewards* according to agent's actions.
- **Goal:** Learn how to take actions to maximize cumulative reward



Characteristics of Reinforcement Learning

- **What makes reinforcement learning different from other machine learning paradigms?**
  - There is no supervisor, only a reward signal
  - Feedback is delayed, not instantaneous
  - Time really matters (sequential, non i.i.d data)
  - Agent's actions affect the subsequent data it receives
- **Examples**
  - Fly stunt maneuvers in a helicopter
  - Defeat the world champion at Backgammon
  - Manage an investment portfolio
  - Control a power station
  - Make a humanoid robot walk
  - Play many different Atari games better than humans

Rewards

- A reward R_t is a scalar feedback signal
- **Indicates** how well agent is doing at step t
- The agent's job is to **maximize** cumulative reward

Reinforcement learning is based on the reward hypothesis:

"All goals can be described by the maximisation of expected cumulative reward."

Examples of Rewards

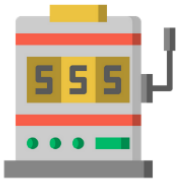
- Fly stunt manoeuvres in a helicopter
  - + reward for following desired trajectory
  - − reward for crashing
- Defeat the world champion at Backgammon
  - +/− reward for winning/losing a game
- Manage an investment portfolio
  - + reward for each $ in bank
- Control a power station
  - + reward for producing power
  - − reward for exceeding safety thresholds
- Make a humanoid robot walk
  - + reward for forward motion
  - − reward for falling over
- Play many different Atari games better than humans
  - +/− reward for increasing/decreasing score

Sequential Decision Making

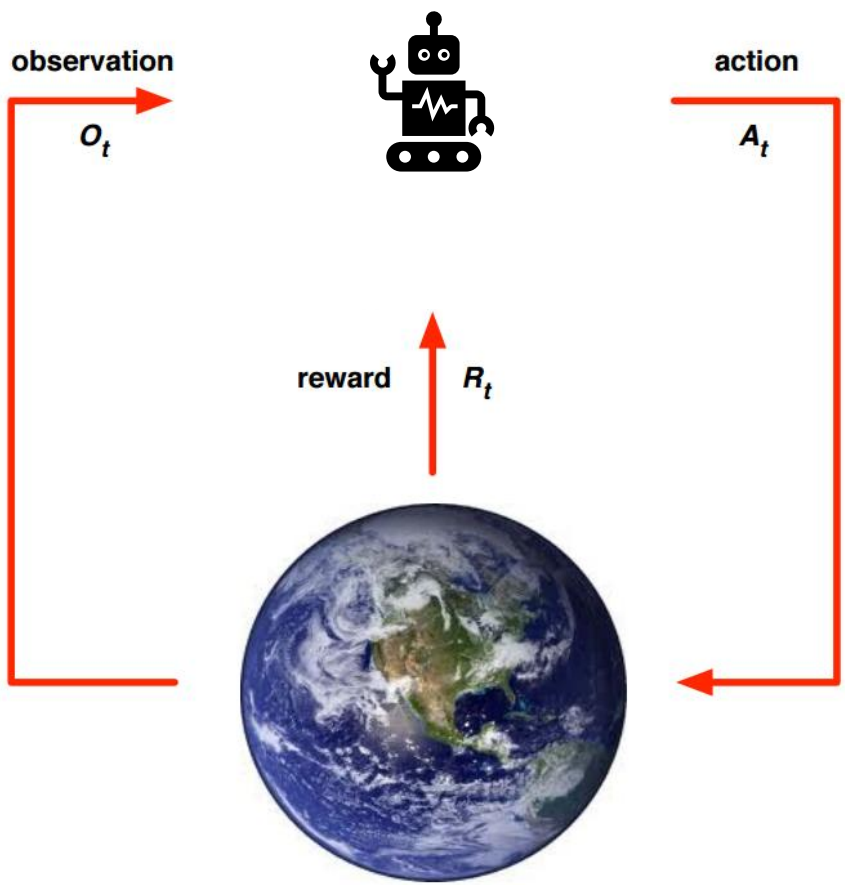
- **Goal**: select actions to maximise total future reward
- **Actions** may have long term consequences
- **Reward** may be delayed
- It may be better to sacrifice immediate reward to gain more long-term reward
- Examples:
  - A financial investment (may take months to mature)
  - Refuelling a helicopter (might prevent a crash in several hours)
  - Blocking opponent moves (might help winning chances many
  - moves from now)

Examples

- Multi-arm bandit

Time	1	2	3	4	…	T
Mean = 5	**5**	5	5	**5**	…	5
 Mean = 6	1	**10**	6	10	…	**10**
Mean = 2.5	2	3	**3**	2	…	2

Agent and Environment



- **At each step t the agent:**
  - Executes action A_t
  - Receives observation O_t
  - Receives scalar reward R_t
- **The environment:**
  - Receives action A_t
  - Emits observation O_{t+1}
  - Emits scalar reward R_{t+1}
- t increments at env. step

History and State

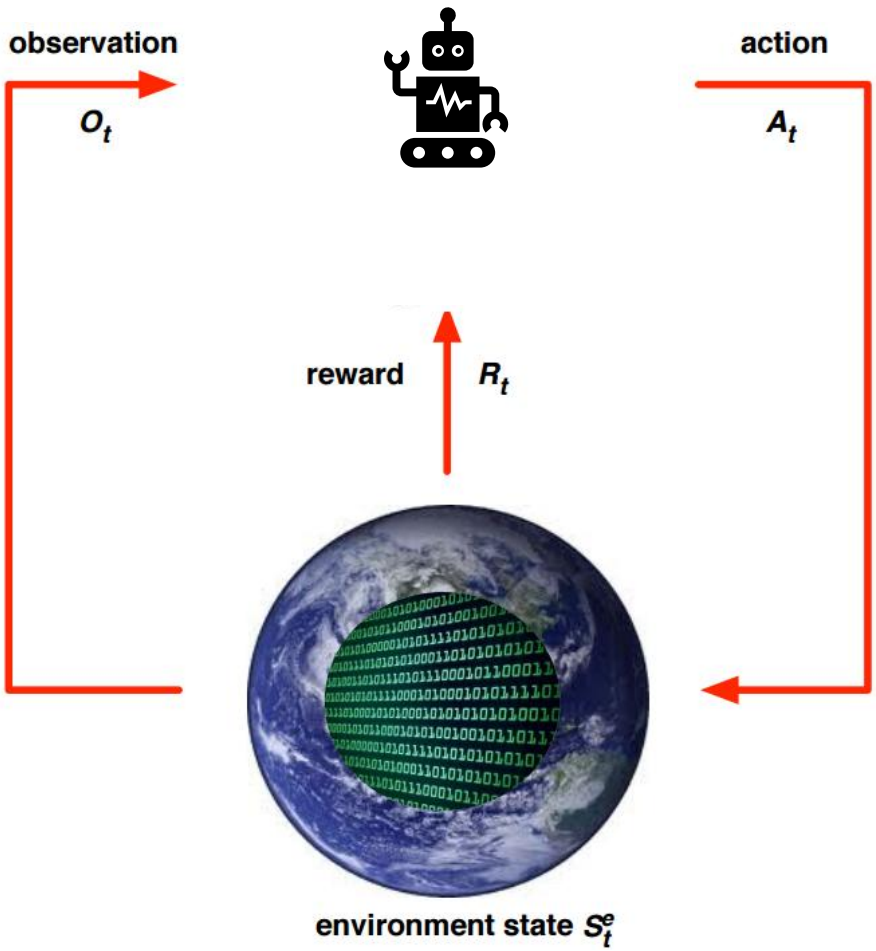
- **The history** is the sequence of observations, actions, rewards, i.e. all observable variables up to time t

$$H_t = O_1, R_1, A_1, \dots, A_{t-1}, O_t, R_t$$

- What happens next depends on the history:
  - The agent selects actions
  - The environment selects observations/rewards
- State is the information used to determine what happens next
- Formally, state is a function of the history:

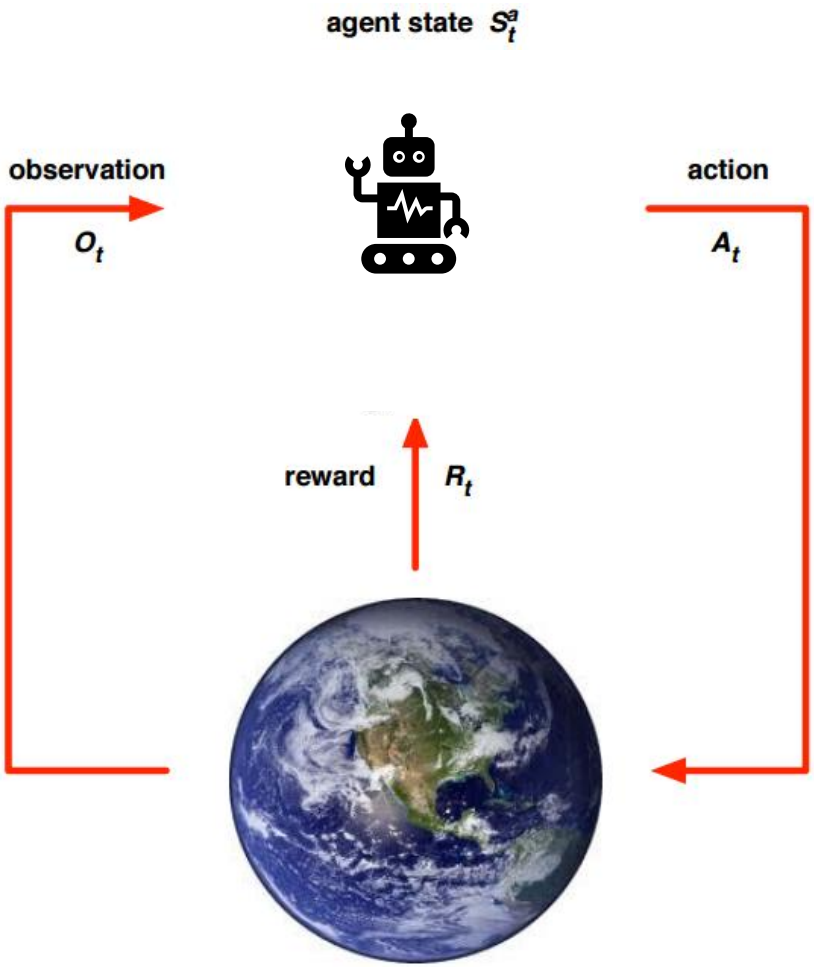
$$S_t = f(H_t)$$

Environment State



- The environment state S_t^e is the environment's private representation
  - i.e., whatever data the environment uses to pick the next observation/reward
- The environment state is not usually visible to the agent
- Even if S_t^e is visible, it may contain irrelevant information.

Agent State



- The agent state S_t^a is the agent's internal representation
  - i.e. whatever information the agent uses to pick the next action
- It is the information used by reinforcement learning algorithms
- It can be any function of history:

$$S_t^a = f(H_t)$$

Information State

- An information state (a.k.a. Markov state) contains all useful information from the history.

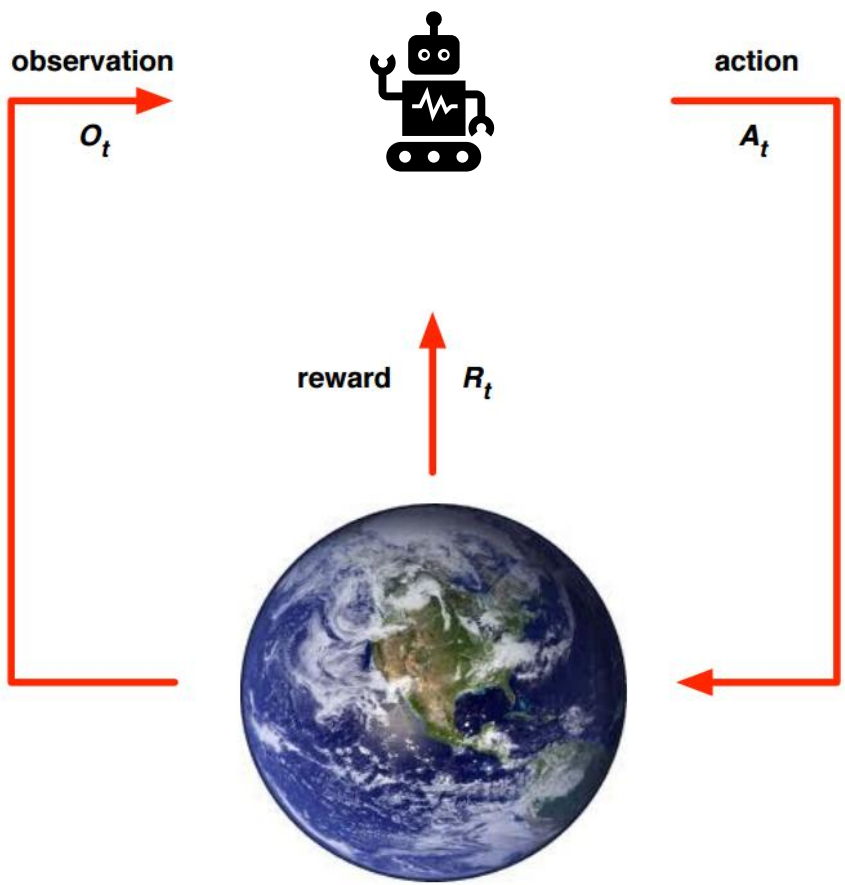
Definition

A state S_t is Markov if and only if

$$\mathbb{P}[\, S_{t+1} | S_t] = \mathbb{P}[S_{t+1} | S1, \dots, S_t]$$

- "The future is independent of the past given the present"
- Once the state is known, the history may be thrown away
  - i.e., the state is a sufficient statistic of the future
- The environment state S_t^e is Markov
- The history H_t is Markov

Fully Observable Environments



- Full observability: agent directly observes environment state

$$O_t = S_t^a = S_t^e$$

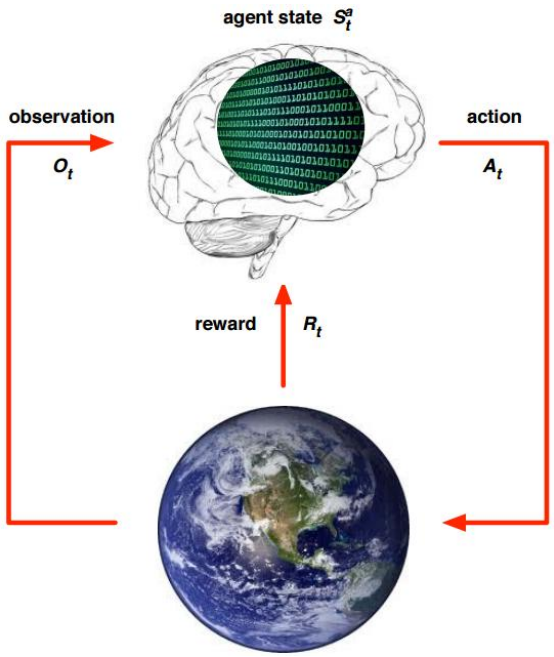
- Agent state = environment state = information state
- Formally, this is a Markov decision process (MDP)

Partially Observable Environments

- Partial observability: agent indirectly observes environment:
  - A robot with camera vision isn't told its absolute location
  - A trading agent only observes current prices
  - A poker playing agent only observes public cards
- Now agent state ≠ environment state
- Formally this is a partially observable Markov decision process (POMDP)
- Agent must construct its own state representation S_t^a
  - Complete history: $S_t^a = H_t$
  - Beliefs of environment state: $S_t^a = (\mathbb{P}[S_t^e = s^1], \dots \mathbb{P}[S_t^e = s^n]$
  - Recurrent neural network: $S_t^a = \sigma(S_{t-1}^a W_s + O_t W_o)$

Major Components of an RL Agent

- An RL agent may include one or more of these components:
  - **Policy**: agent's behaviour function
  - **Value function**: how good is each state and/or action
  - **Model**: agent's representation of the environment



Policy

- A policy is the agent's behaviour
- It is a map from state to action
- **Deterministic policy**

$$a = \pi(s)$$

- **Stochastic policy**

$$\pi(a|s) = \mathbb{P}[\mathrm{A_t} = \mathrm{a}|\mathrm{S_t} = \mathrm{s}]$$

Value Function

- Value function is a prediction of future reward
- Used to evaluate the goodness/badness of states
- And therefore, to select between actions, e.g.

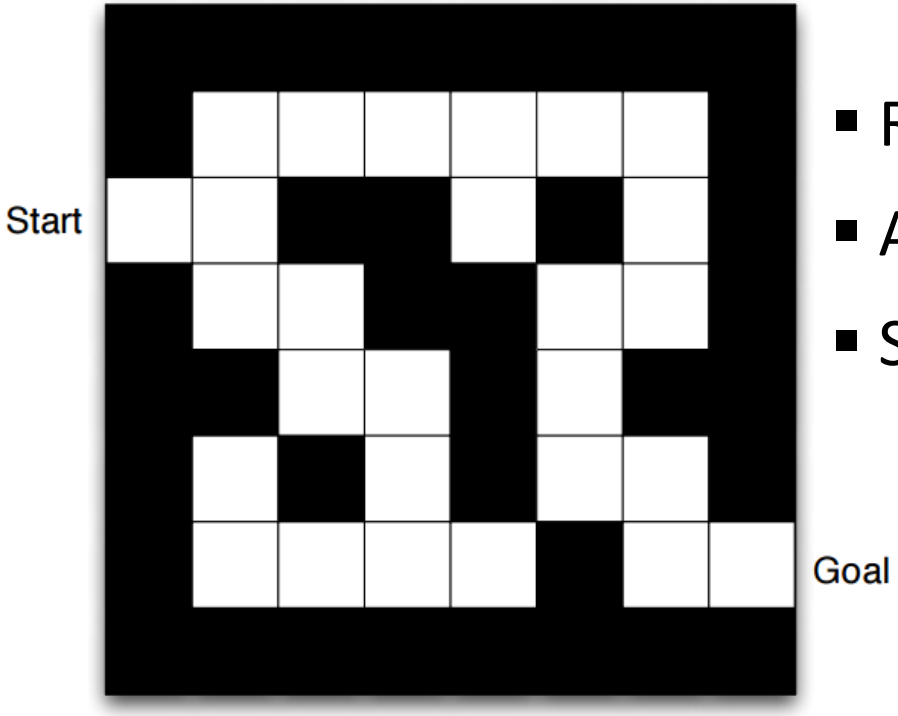
$$v_\pi(s) = \mathbb{E}_\pi \left[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots \mid S_t = s \right]$$

Model

- A model predicts what the environment will do next
- $\mathcal{P}$ predicts the next state
- $\mathcal{R}$ predicts the next (immediate) reward, e.g

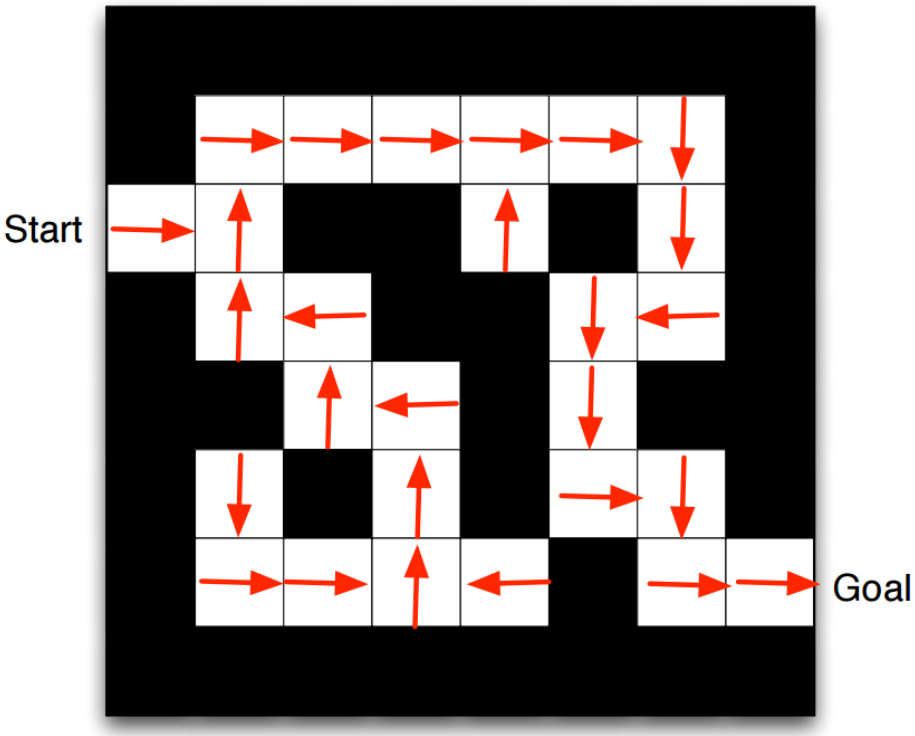
$$\mathcal{P}_{ss'}^{a} = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a]$$
$$\mathcal{R}_{s}^{a} = \mathbb{E}\left[R_{t+1} \mid S_t = s, A_t = a\right]$$

Maze Example



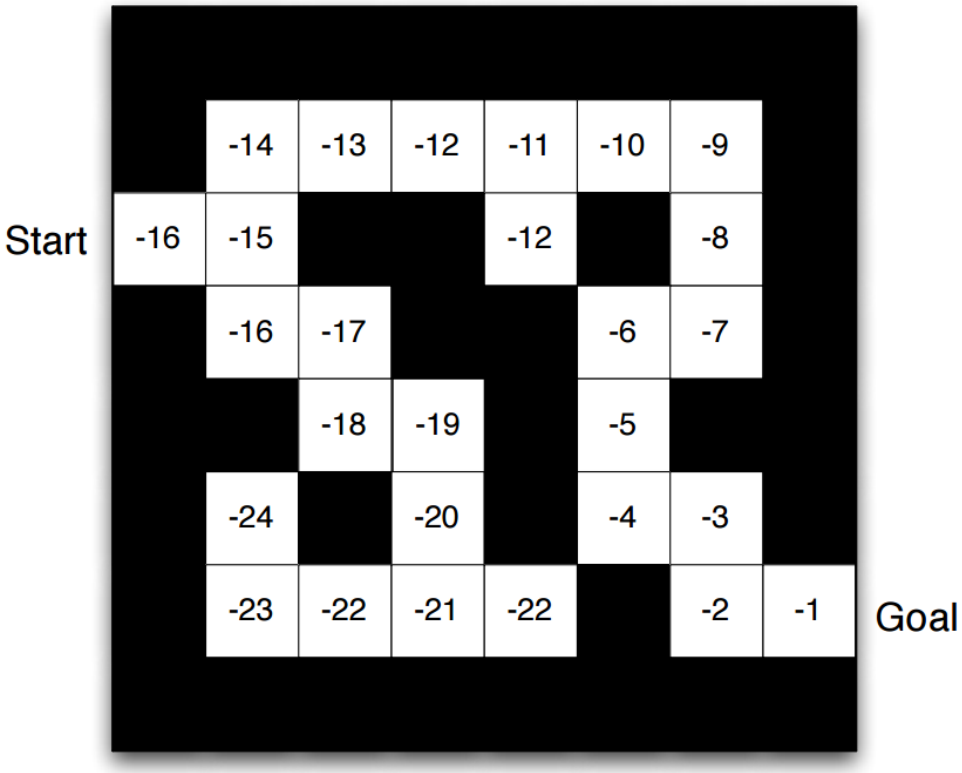
- Rewards: -1 per time-step
- Actions: N, E, S, W
- States: Agent's location

Maze Example: Policy



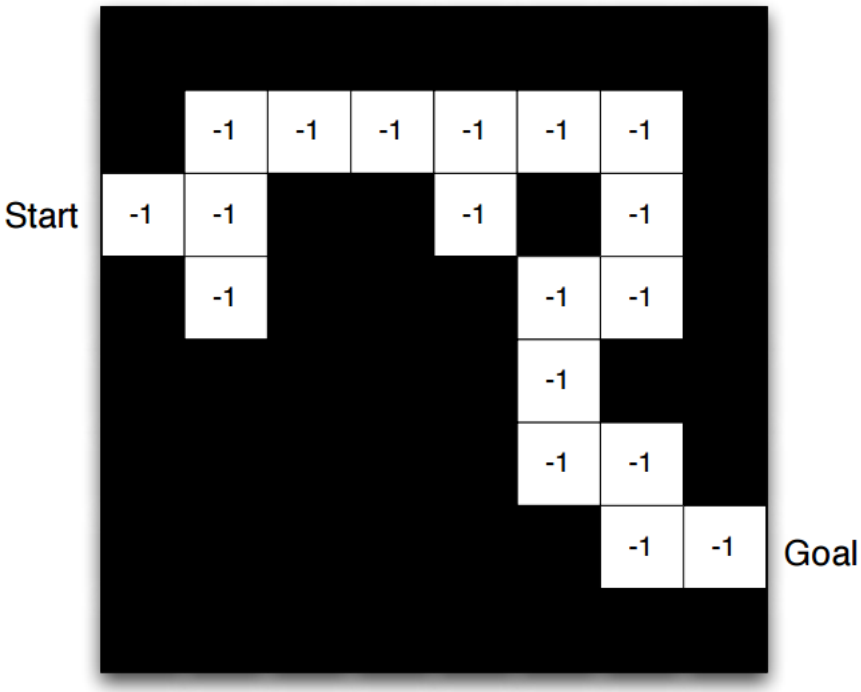
- Arrows represent policy π(s) for each state s

Maze Example: Value Function



- Numbers represent value $v_\pi(s)$ of each state s

Maze Example: Model



- Agent may have an internal model of the environment
- Dynamics: how actions change the state
- Rewards: how much reward from each state
- The model may be imperfect

- Grid layout represents transition model $\mathcal{P}_{ss'}^{a}$,
- Numbers represent immediate reward $\mathcal{R}_{s}^{a}$ from each state s (same for all a)

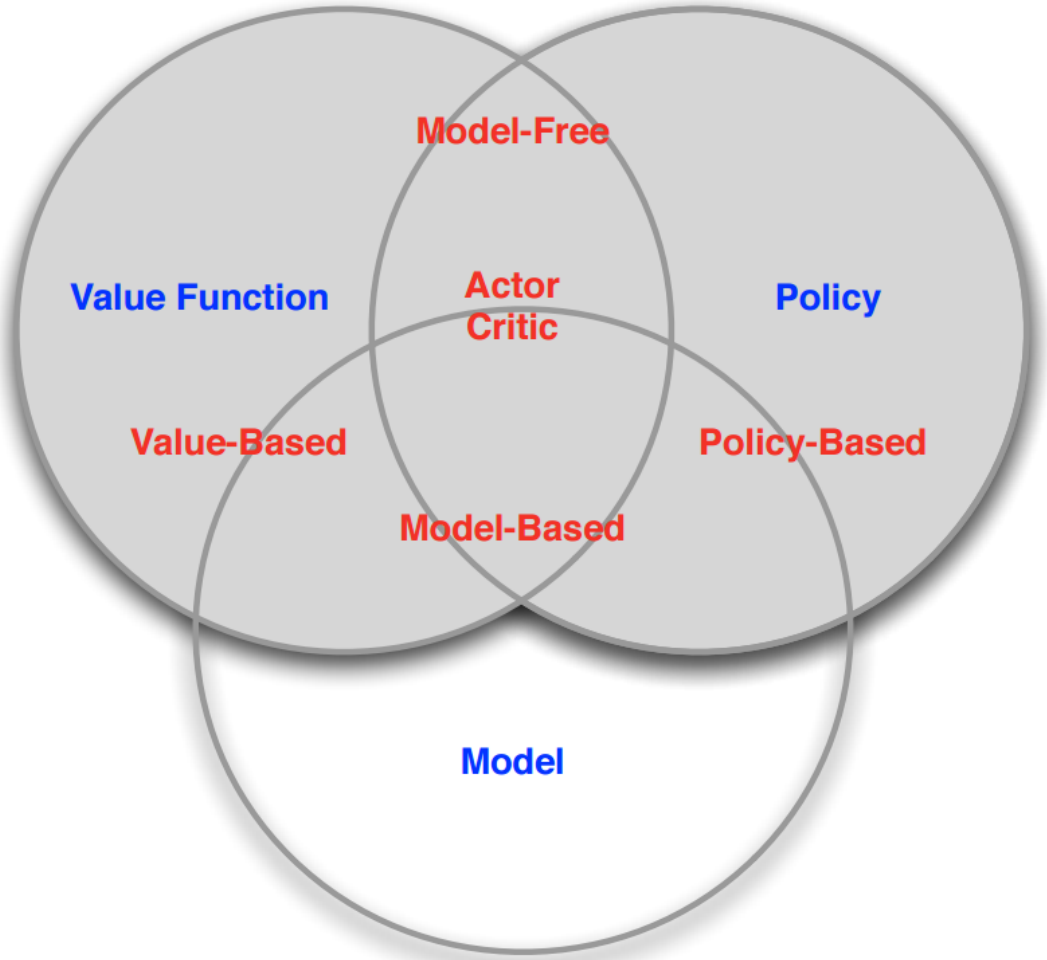
Categorizing RL agents

- **Value Based**
  - No Policy (Implicit)
  - Value Function
- **Policy Based**
  - Policy
  - No Value Function
- **Actor Critic**
  - Policy
  - Value Function

Categorizing RL agents (2)

- **Model Free**
  - Policy and/or Value Function
  - No Model
- **Model Based**
  - Policy and/or Value Function
  - Model

RL Agent Taxonomy

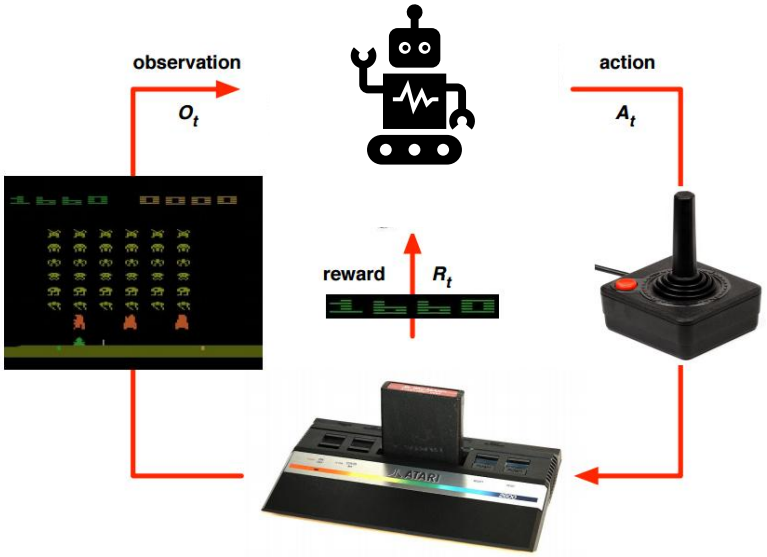


Learning and Planning

Two fundamental problems in sequential decision making

- **Reinforcement Learning:**
  - The environment is initially unknown
  - The agent interacts with the environment
  - The agent improves its policy
- **Planning:**
  - A model of the environment is known
  - The agent performs computations with its model (without any external interaction)
  - The agent improves its policy
    - a.k.a. deliberation, reasoning, introspection, pondering, thought, search

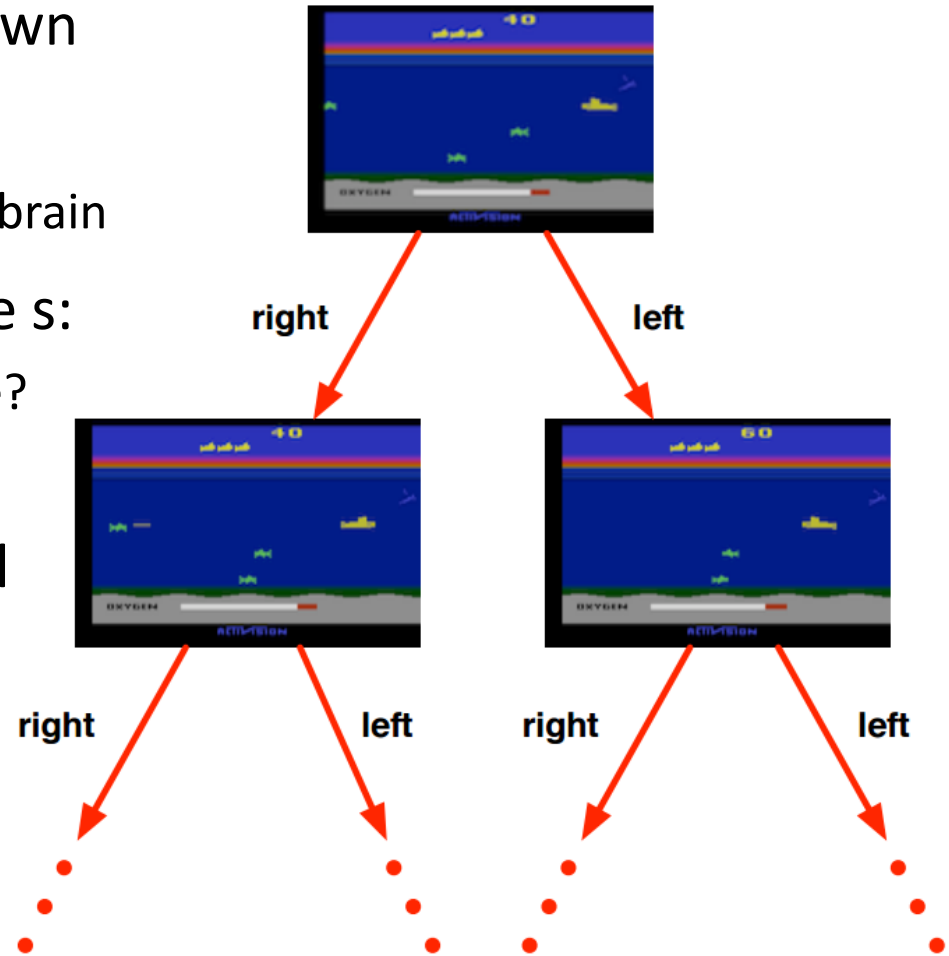
Atari Example: Reinforcement Learning



- Rules of the game are unknown
- Learn directly from interactive game-play
- Pick actions on joystick, see pixels and scores

Atari Example: Planning

- Rules of the game are known
- Can query emulator
  - perfect model inside agent's brain
- If I take action a from state s:
  - what would the next state be?
  - what would the score be?
- Plan ahead to find optimal policy
  - e.g. tree search



Exploration and Exploitation

- Reinforcement learning is like trial-and-error learning
- The agent should discover a good policy
- From its experiences of the environment
- Without losing too much reward along the way
- **Exploration** finds more information about the environment
- **Exploitation** exploits known information to maximise reward
- It is usually important to explore as well as exploit

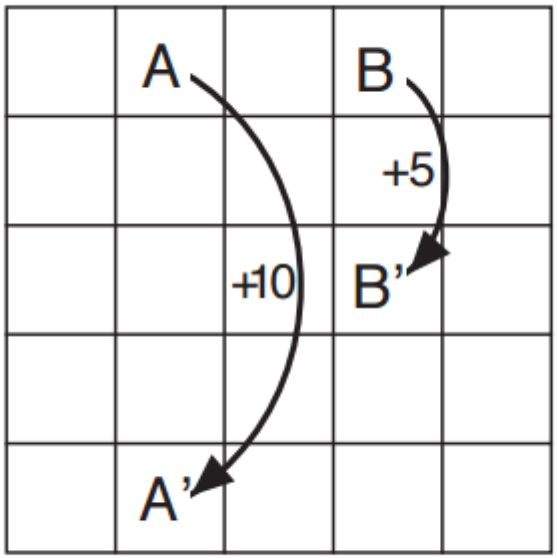
Examples

- Restaurant Selection
  - Exploitation Go to your favourite restaurant
  - Exploration Try a new restaurant
- Online Banner Advertisements
  - Exploitation Show the most successful advert
  - Exploration Show a different advert
- Oil Drilling
  - Exploitation Drill at the best-known location
  - Exploration Drill at a new location
- Game Playing
  - Exploitation Play the move you believe is best
  - Exploration Play an experimental move

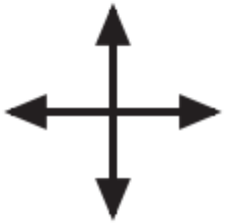
Prediction and Control

- **Prediction**: evaluate the future
  - Given a policy
- **Control**: optimise the future
  - Find the best policy

Gridworld Example: Prediction



(a)



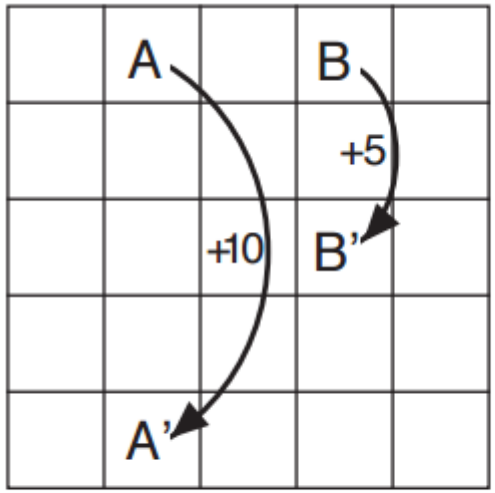
Actions

3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

(b)

- What is the value function for the uniform random policy?

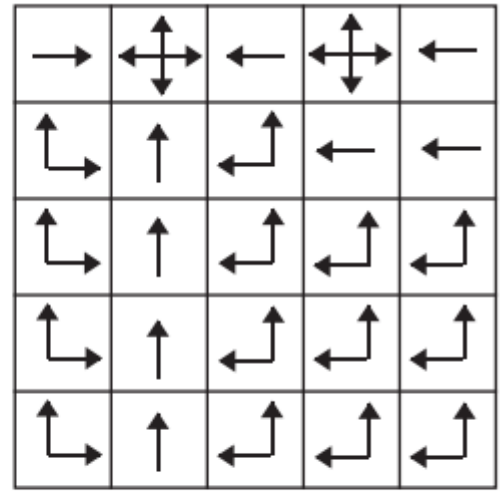
Gridworld Example: Control



a) gridworld

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

b) v_*



c) π_*

- What is the optimal value function over all possible policies?
- What is the optimal policy?

Multi-Agent RL

Multi-Agent RL

Challenges

- **Sample Efficiency:** RL algorithms often require a large number of interactions with the environment to learn effective policies, which can be impractical in real-world scenarios where data collection is costly or time-consuming.
- **Reward Design:** Designing appropriate reward signals that lead to desired behaviors without unintended consequences is challenging. Poorly designed rewards can lead to suboptimal or harmful behaviors.
- **Partial Observability:** In many real-world scenarios, the agent does not have access to the full state of the environment, making it difficult to make informed decisions.
- **Multi-Agent Environments**: Coordinating multiple agents that learn simultaneously introduces additional complexity, as each agent's learning process can affect the others.
- **Safety and Ethics**: Ensuring that RL agents behave safely and ethically, especially in critical applications, is a growing concern. This includes avoiding harmful exploration and ensuring compliance with ethical guidelines.

Conclusion

- **Main Takeaways**
  - RL involves an agent learning to make decisions through trial and error in an environment, aiming to maximize cumulative rewards.
  - Key Components: agent, environment, states, actions, rewards, and policy.
  - Exploration vs. Exploitation
  - Challenges